Sustainability Economics: Hermit Economics

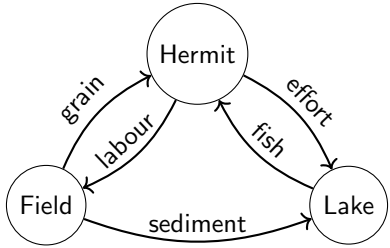
John Janmaat

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Hermit System

- Hermit,
  - The social element of the social ecological system.
  - Contributes labour and consumes the fruits of that labour.
- Field,
  - One ecological element of the system.
  - Produces grain, with labour as the main input.
- Lake.
  - One ecological element of the system.
  - Produces fish, with fishing effort (time spent fishing) as the main input.

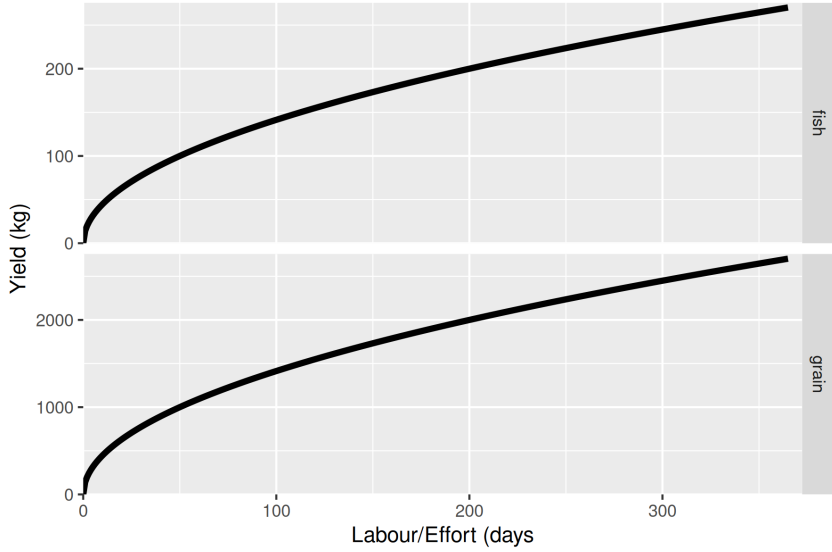
Hermit System



Production Functions

- A *technology* is a way that inputs are converted to outputs.
- A *production function* is a way of representing the technology.
  - The production function is a simplification.
  - The technology behaves the way that physics, chemistry, etc. require it to.
  - All models are wrong, some are useful.
- Three measures are commonly used to describe production at each input level.
  - *Total Product* is the total output produced given a particular input level.
  - *Average Product* is the total product divided by the input level.
  - *Marginal Product* is how much the total output changes if the level of input is changed by a small amount.

Production Functions



Total, Average and Marginal Product

- Total Product (TP) and Average Product (AP) are based on a level of input.

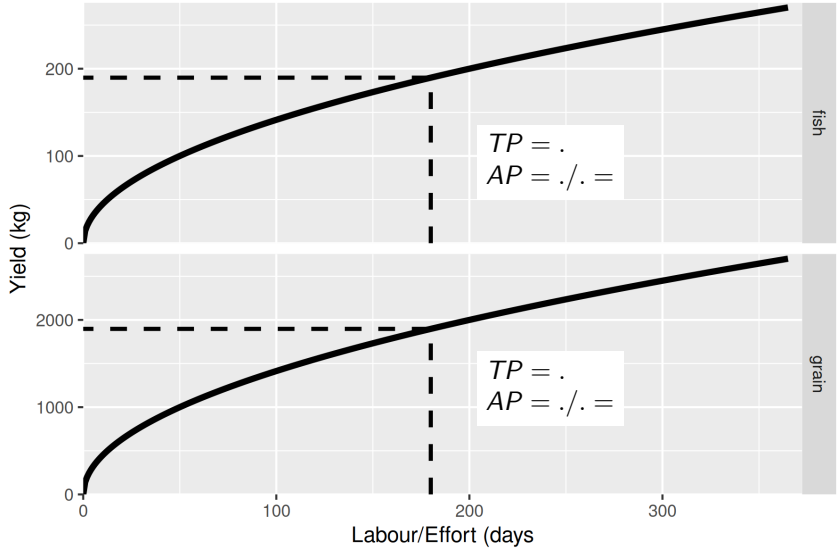
$$AP = TP/L$$

  - L can be any input, not just labour.
- Marginal Product (MP) is based on a small change in in the level of an input, around a specific level of that input.

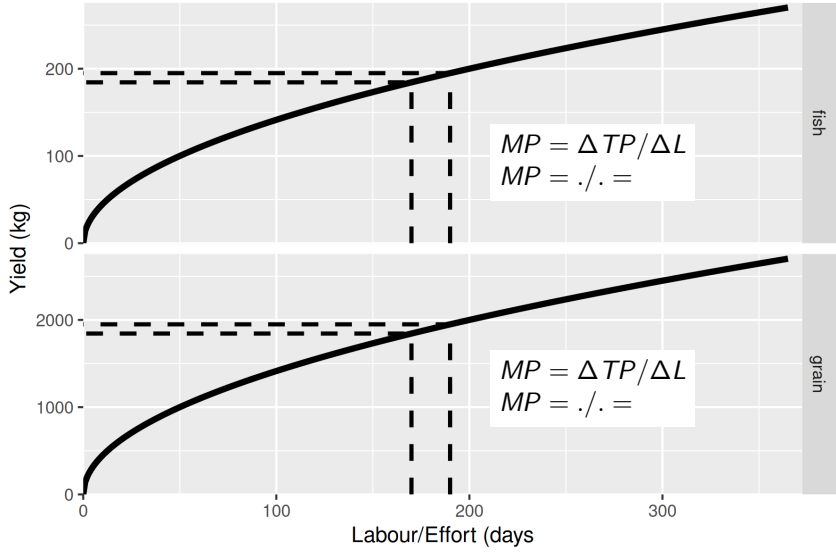
$$MP = \Delta TP/\Delta L = \partial TP/\partial L$$

  - The marginal product measures how much extra output would be produced if a small amount of extra input is used.
  - It is the slope of a line that just touches the production function at the input level.

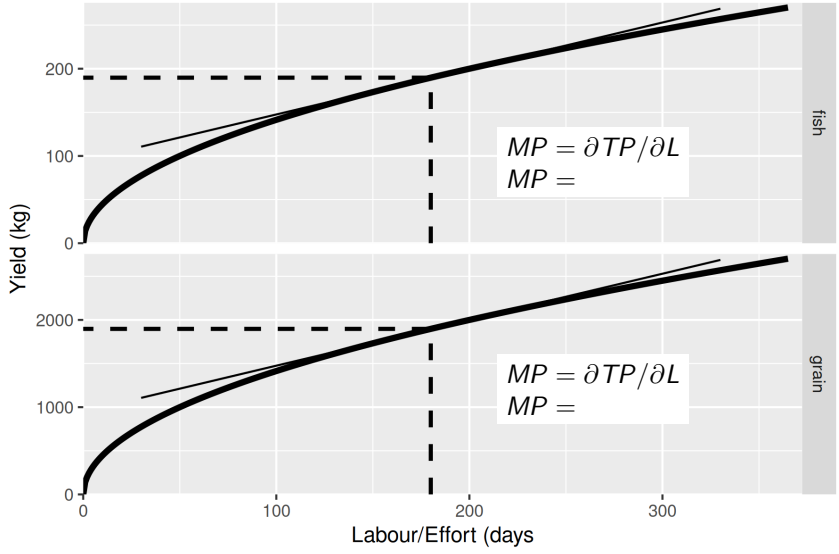
Total, Average and Marginal Product



Total, Average and Marginal Product



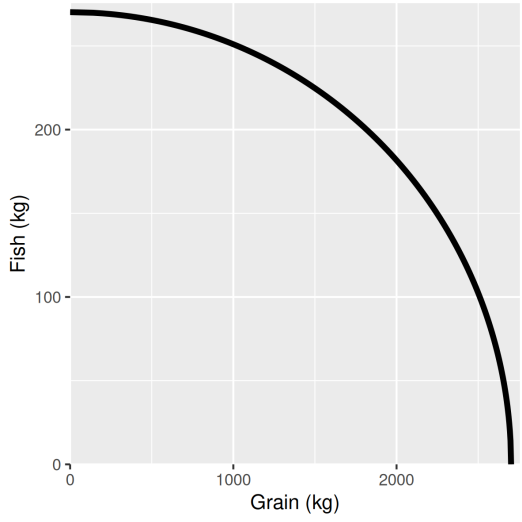
Total, Average and Marginal Product



Production Possibilities Frontier

- The hermit has 365 days per year to either work in the grain field or catch fish.
- Spending more time working in the grain field means less time catching fish.
- The *opportunity cost* of the grain produced by spending one more day in the field is the fish that would have been caught if that day had been spent catching fish.
- The **opportunity cost** of a choice is the value of the next best alternative foregone.

Production Possibilities Frontier



Production Possibilities Frontier

